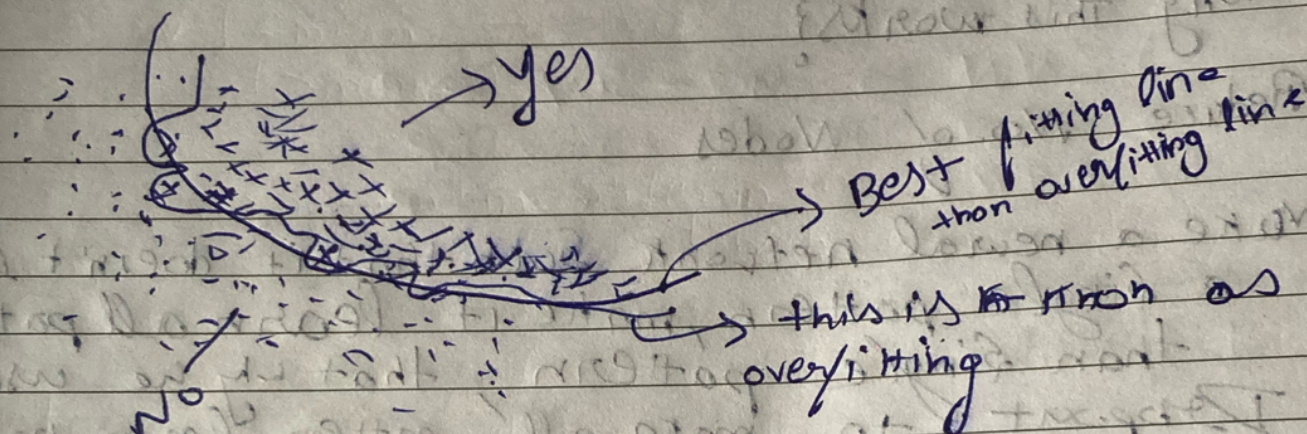


Dropouts in ANN

Overfitting - model best on training data poor on unseen data

Dropout is a method to reduce overfitting in neural network & improve Neural Networks



We can solve this by

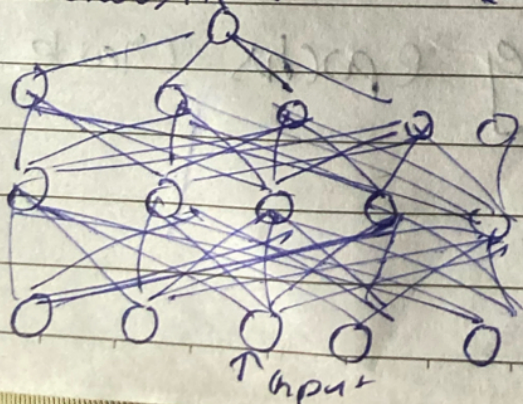
- Adding more data
- Reduce complexity of your layer
- Early Stopping
- Regularization
- Dropout

There in evy epoch random nodes are disabled based on dropout rate

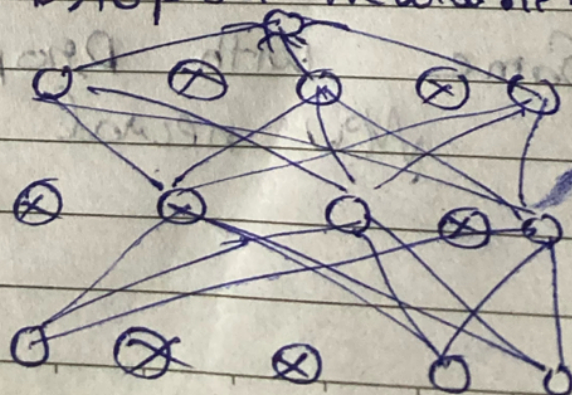
$p=0.5$
So: nodes are disabled

This is a neural network which occurs overfitting

Neural Network



Dropout Neural Network



In every epoch we randomly decide which nodes will be ON/OFF

So in every we train on different Neural Networks
Around 2% Accuracy improved using Dropout

Why this works?

Reduce No. of Nodes

Make a neural network such that it doesn't learn a single pattern, make it learn all patterns than single pattern, that's why we use Dropout to make all pattern learn not a single pattern focus

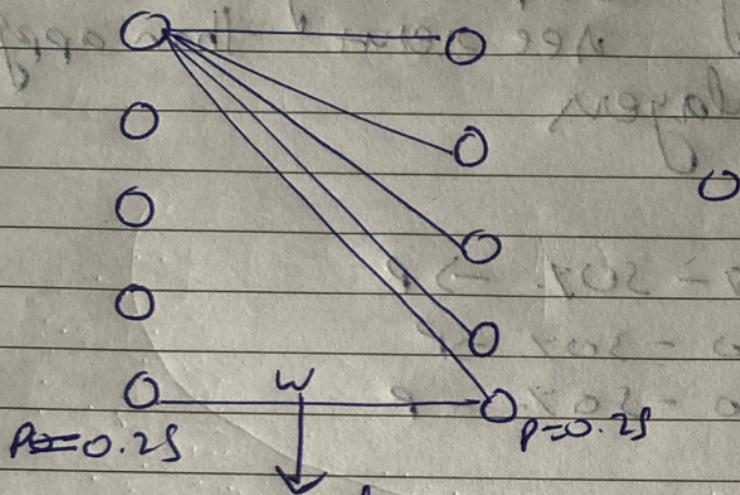
Random Forest Analogy

↳ Dekh random forest bhi same trick use krta. vo prediction k liye w numbers of Decision Tree bnata h o highest probability wali wala yes or No output deta h

Same with Dropout every epochs create new neural network

How prediction work?

↳ The Dropout is applied only at time of training not Testing



let the weight while testing-training be w so what weight will be while testing?

$$= w(1 - 0.25) = w \times 0.75$$

from sklearn.tensorflow.keras.layers import Dropout

If p value (value of Dropout) is very less than ~~over~~ overfitting occurs if ~~th~~ it is greater than under fitting

$$0.2 \leq p \leq 0.5$$

↳ Best p value



Tips & Tricks

- 1) Overfitting $\uparrow$, Underfitting $\downarrow$
- 2) first apply dropout only on last layer see result then apply on other layers

- 3) CNN $\rightarrow$ 40-50% $\rightarrow$ P
- 4) RNN $\rightarrow$ 20-30% $\rightarrow$ P
- 5) ANN $\rightarrow$ 10-50% $\rightarrow$ P

Drawbacks

- 1) convergence $\rightarrow$ delay / slow
- 2) cost function value changes